Does Lifestyle Information Improve Hypertension Prediction? A Leakage-Controlled Evaluation on NHANES

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Abstract

Does readily available lifestyle information improve machine-learning prediction of hypertension beyond routine demographic and anthropometric data? On NHANES 2017–March 2020, five model families (Logistic Regression, Random Forest, Gradient Boosting, Bagged Decision Trees, and XGBoost) were evaluated with fold-local preprocessing and paired bootstrap confidence intervals. Adding physical activity, alcohol use, smoking, and sleep information showed no consistent detectable improvement in held-out discrimination: every paired ROC-AUC confidence interval included zero, including under a stricter measured-blood-pressure phenotype and without synthetic oversampling. Mean cross-model SHAP rank agreement fell from 0.873 to 0.806 over all encoded columns, but this decline was largely attenuated over the 19 baseline columns shared by both arms (0.873 vs. 0.863), and likewise under both robustness checks; attribution-agreement comparisons across feature sets of different sizes therefore require a matched comparison space. In exploratory analysis of untreated adults (measured-blood-pressure outcome), all five models showed small positive differences, two with unadjusted intervals excluding zero, meriting confirmation. These results do not imply that lifestyle factors are clinically unimportant; within this cohort, the evaluated lifestyle block added little incremental discrimination beyond an already-informative baseline.

Keywords: hypertension prediction, NHANES, cross-validation, model interpretability, SHAP, class imbalance

Data and Code Availability This study uses publicly available, de-identified NHANES 2017–March 2020 data from NCHS/CDC, freely accessible at https://wwwn.cdc.gov/nchs/nhanes/continuousnhanes/default.aspx?Cycle=2017-2020. All code, result tables, and figures are publicly available at https://github.com/urielbenymon/nhanes-lifestyle-hypertension, including scripts that download the NHANES data directly from CDC and reproduce every reported analysis.

Institutional Review Board (IRB) This study is a secondary analysis of publicly available, de-identified NHANES data; no new data were collected and no interaction with human subjects occurred. No IRB review was sought for this analysis.

1. Introduction

Hypertension affects a large share of the global adult population and remains a leading modifiable contributor to cardiovascular disease, with sustained increases in prevalence and shifting burden documented across more than a hundred million participants worldwide (Zhou et al., 2021). The 2017 ACC/AHA guideline lowered the diagnostic threshold to 130/80 mmHg, expanding the population classified as hypertensive and increasing interest in identifying who is at risk from readily available information (Whelton et al., 2018). Physical activity, alcohol use, smoking, and sleep are frequently recommended targets for hypertension prevention and management, making it practically useful to know how much they add to prediction built from routinely collected demographic and anthropometric data.

A substantial body of work applies machine learning to hypertension prediction, spanning multiple model families, feature sets, and evaluation designs (Silva et al., 2022). Closest to the present setting, Potvin et al. (2025) predict hypertension from NHANES demographic and lifestyle variables using Logistic Regression, Random Forest, Gradient Boosting, and Bagged Decision Trees with a single 90/10 evaluation split; Huang and Huang (2023) predict hypertension from the same NHANES 2017–March 2020 cycle using XGBoost and SHAP over demographic and nutritional covariates; and López-Martínez et al. (2020) predict hypertension from an earlier NHANES

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cycle using a neural network. Collectively, this work establishes that demographic, anthropometric, clinical, nutritional, and behavioral information can each participate in hypertension prediction, though evaluation designs, phenotype definitions, and cohorts vary considerably across studies, which complicates direct comparison of reported performance.

Physical activity, alcohol use, smoking, and sleep are each independently associated with hypertension risk (Barone Gibbs et al., 2021; Hosseini et al., 2024; Cecchini et al., 2024; Virdis et al., 2010). However, established risk-factor association is a distinct question from incremental predictive value: a variable can be epidemiologically relevant to hypertension while adding little additional discriminative information once other readily available predictors are already present. It remains less clear how much a focused, modifiable lifestyle block contributes beyond an already-informative demographic and anthropometric baseline under leakage-controlled evaluation, and whether expanding the feature space in this way changes how consistently different model families identify important predictors, a question distinct from whether any single model's attribution is trustworthy in isolation.

This gap is addressed using the NHANES 2017–March 2020 cycle, comparing a demographic/SES-and-anthropometric baseline against a lifestyle-augmented arm across five model families (Section 2). The first empirical question (EQ1) is whether the lifestyle block provides incremental held-out predictive value beyond the baseline. The second empirical question (EQ2) is whether it changes cross-model agreement in SHAP feature importance, and whether any change persists when both arms are compared over the same baseline columns rather than over feature spaces of different sizes. Robustness is tested under an alternative hypertension phenotype and removal of synthetic oversampling (Appendix E), supplemented by calibration and decision-curve analysis, feature-subset ablations, training-seed and repeated-split variability, an untreated-adult subgroup, and lifestyle–baseline redundancy diagnostics.

Relative to Potvin et al. (2025), the contribution lies mainly in the evaluation design: an explicit incremental-value comparison with paired feature arms on the same participants and split; fold-local preprocessing, including SMOTE, with cross-validated tuning; paired held-out uncertainty for the lifestyle-minus-baseline difference; XGBoost as a fifth model family; and a cross-model attribution analysis with robustness checks. Two further findings emerge. First, cross-model SHAP agreement compared across feature sets of different sizes is sensitive to how the comparison space is defined, so such comparisons require a matched comparison space. Second, the full-cohort null result may not hold in every subpopulation: in an exploratory analysis of untreated adults, all five models showed small positive lifestyle-minus-baseline AUC differences, a result presented as hypothesis-generating.

2. Methods

Cohort and outcome. This study uses the NHANES 2017–March 2020 pre-pandemic release, restricted to adults aged 18 and older who were both interviewed and examined. Of 15,560 initial participants, 8,965 met this criterion, and 8,024 additionally had a valid oscillometric blood pressure reading and were phenotype-eligible (Table 1). Hypertension was defined using the 130/80 mmHg threshold established by Whelton et al. (2018), averaged across up to three readings per participant. The primary outcome, Phenotype B, is positive if measured blood pressure meets this threshold or the participant reports currently taking antihypertensive medication (4,311/8,024, 53.7%); the medication clause is not redundant with measured BP alone, since 903 of these positives have normal measured blood pressure. A sensitivity analysis, Phenotype A, uses the measured-BP criterion alone (3,408/8,024, 42.5%). Full phenotype definitions, including a broader self-report-inclusive Phenotype C, and the complete predictor dictionary appear in Appendix A. Participants were split once into a fixed 90% development set (7,221) and a 10% held-out test set (803), stratified on Phenotype B; this split was reused unchanged for the Phenotype A analysis so that only the outcome label, not cohort composition, varied. Survey design variables were retained but not used as predictors, so reported estimates describe predictive performance within this analytic cohort rather than population-representative prevalence.

Features and modeling. Two feature arms are compared using the same participants and split: a *baseline* arm (demographic/SES and anthropometric predictors) and a *lifestyle-augmented* arm that adds physical activity, alcohol use, smoking, and sleep. This block was prespecified as commonly recommended modifiable behaviors from the guideline and meta-analytic evidence cited in Section 1, rather than

selected using the outcome, which would tune it to the data it is tested on and bias the incremental comparison; every candidate variable has a documented inclusion or exclusion decision, made from the study aims and cited literature without formal clinical expert review (Appendix A). Twenty-six raw lifestyle items were consolidated into 11 derived predictors using skip-pattern-aware rules. This pairing is the instrument for the first empirical question: whether the lifestyle block adds incremental predictive value beyond an already-informative baseline. Each arm was evaluated with five model families: Logistic Regression, Random Forest, Gradient Boosting, Bagged Decision Trees, and XGBoost (Chen and Guestrin, 2016). Every pipeline applies one-hot encoding, median imputation, standardization for Logistic Regression only, and SMOTE (Chawla et al., 2002), all fit inside an `imbalanced-learn` pipeline (Lemaitre et al., 2016) so that no learned transformation, including SMOTE's oversampling, was ever fit on validation or test data, addressing a documented source of leakage-driven irreproducibility in machine-learning-based research (Kapoor and Narayanan, 2023). Hyperparameters were selected via 5-fold stratified cross-validation (Kohavi, 1995) on the 90% development set only, scored by ROC-AUC: an exhaustive grid for Logistic Regression, and 25-candidate randomized search (Bergstra and Bengio, 2012) over tree depth, ensemble size, and related regularization parameters for the remaining four models, run independently per model per feature arm, ten configurations total, with full search spaces and selected values in Appendix B. This is a single cross-validation layer, not a nested search. Selected classifier hyperparameters were frozen and reused, unmodified, for held-out evaluation, interpretability analysis, both robustness checks, and all supplementary analyses below.

Evaluation. ROC-AUC and PR-AUC are reported. Held-out performance uses bootstrap 95% confidence intervals (Efron and Tibshirani, 1993) on the fixed 10% test partition. In the main-model evaluation, the test set was examined twice: once with library-default hyperparameters and once after all hyperparameters were frozen; it was never used to fit models or select hyperparameter values. The default results contributed to the decision to tune, but that decision was supported by development data alone: XGBoost's default cross-validated ROC-AUC (0.782 baseline, 0.785 lifestyle-augmented) was already well below that of Logistic Regression (0.813, 0.815). All five models were then tuned with the same development-set criterion, not XGBoost alone, and no search space, feature set, or preprocessing decision was changed in response to held-out results. The default run had already produced the same null result for the first question (all five paired ROC-AUC intervals included zero; point estimates -0.014 to $+0.010$). The final test evaluation is nonetheless not fully independent of earlier modeling decisions, and the supplementary analyses below reuse the same test set.

To test the first question, a paired bootstrap confidence interval is computed for the lifestyle-minus-baseline difference, resampling the same test participants for both arms each draw (Figure 1; full values in Appendix C). For the second question, whether adding lifestyle information changes how consistently model families attribute importance, SHAP values are computed (Lundberg and Lee, 2017) using `TreeExplainer` (Lundberg et al., 2020) for Random Forest, Gradient Boosting, and XGBoost, and a model-agnostic explainer for Bagged Decision Trees, which `TreeExplainer` does not support; these describe learned attribution, not a causal claim about hypertension risk. Following Krishna et al. (2022)'s framing of cross-model explanation disagreement, agreement is quantified two ways: top-10 Jaccard overlap and Spearman rank correlation over the complete ordering, both over the post-encoding feature space and, because the two arms rank different numbers of columns (19 vs. 36), also over the 19 baseline columns they share (Table 2; Appendix D). Robustness is tested by refitting under Phenotype A and by refitting with SMOTE removed, holding classifier hyperparameters fixed (Appendix E).

Supplementary analyses, all using the frozen hyperparameters, comprise SHAP agreement controls (matched comparison space, a shuffled-lifestyle control, and predictor- and domain-level aggregation; Appendix F); calibration and exploratory decision-curve analysis (Appendix G); lifestyle–baseline redundancy diagnostics on the development set (Appendix H); an untreated-adult subgroup (Appendix I); feature-subset ablations (Appendix J); and training-seed and repeated-split variability (Appendix K). Reproducibility and ethics details appear in Appendices L and M.

3. Results

Table 1 summarizes the analytic cohort and both feature arms. After cross-validated hyperparameter selection, lifestyle augmentation produced only small,

Table 1: Cohort construction, phenotype definitions, and feature-arm composition. Phenotype C is not analyzed in the main text; see Appendix A.

Item	**Value**
Initial NHANES 2017–March 2020 participants	15,560
Interviewed and MEC-examined	14,300
Age $\geq$ 18	8,965
Phenotype-eligible (valid BP reading)	8,024
Primary Phenotype B (measured $\geq$130/80 or medication)	4,311 (53.7%)
Sensitivity Phenotype A (measured only)	3,408 (42.5%)
Development / held-out test split	7,221 / 803
Baseline predictors	Demographic/SES + anthropometric
Lifestyle-augmented predictors	Baseline + activity, alcohol, smoking, sleep

inconsistent changes in development-set discrimination across the five model families, with ROC-AUC differences ranging from −0.0005 for Random Forest to +0.0035 for Gradient Boosting; Random Forest is the only model with a negative development-set delta. These are development/model-selection estimates rather than an independent confirmation, since the same cross-validation folds informed hyperparameter selection.

On the fixed held-out test set, lifestyle augmentation did not reliably improve discrimination for any model family (Figure 1). Paired bootstrap ROC-AUC differences ranged from −0.0028 for XGBoost to +0.0002 for Bagged Decision Trees, and all five 95% confidence intervals included zero; the corresponding PR-AUC differences showed the same pattern, with full values in Appendix C. Small development-set differences did not translate into reliable held-out improvement. This result addresses incremental predictive discrimination specifically, within this analytic cohort; it does not speak to whether lifestyle factors are associated with hypertension more broadly.

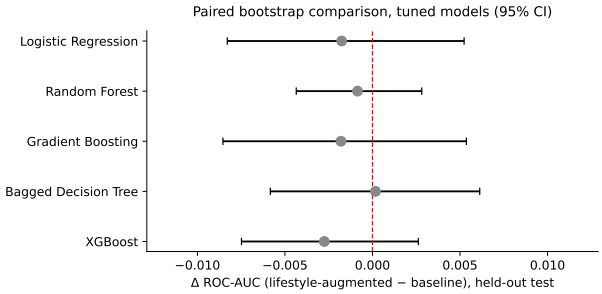


Figure 1: Paired bootstrap comparison of held-out ROC-AUC, lifestyle-augmented minus baseline, tuned models (95% CI). All five intervals include zero.

At the encoded-column level, mean pairwise cross-model SHAP rank agreement decreased after lifestyle augmentation (Spearman 0.873 to 0.806; Table 2). However, the decrease was substantially attenuated when both arms were compared on the same 19 baseline columns (0.873 vs. 0.863; paired bootstrap $\Delta = -0.010$, 95% CI $[-0.021, 0.010]$). A shuffled-lifestyle control, which preserves the 36-column feature space and each lifestyle predictor's marginal distribution while disrupting its observed associations, showed a similar full-column decline (0.776) with similar shared-column agreement (0.863). Top-10 Jaccard showed a smaller lifestyle-specific component: it fell further with real lifestyle columns (0.699) than with shuffled ones (0.745), as real lifestyle features such as trouble sleeping entered some models' top 10 but not others. Shuffled columns received 5.3% of total mean |SHAP|, compared with 11.3% for real lifestyle columns. Predictor- and domain-level analyses indicated substantial agreement at coarser resolutions, subject to their aggregation and unequal-list-length limitations: summed within six domains, mean pairwise Spearman agreement in the lifestyle arm was 0.903, and every model ranked demographic/SES first and anthropometric second (Appendix F). The full-column comparison is therefore

sensitive to how the feature space is defined, and it should not be interpreted as a general deterioration in cross-model explanation agreement. All measures describe each model's learned attribution, not a causal ranking of hypertension risk factors, and are interpreted descriptively; an illustrative single-model predictor-level ranking appears in Appendix D.

The absence of a reliable held-out lifestyle benefit persisted under two robustness checks (Appendix E). Refitting the frozen configurations under the measured-BP-only Phenotype A left all five paired confidence intervals including zero, despite Phenotype A's substantially lower overall discrimination (development-set ROC-AUC near 0.71–0.72, versus approximately 0.81 for the primary phenotype). Removing SMOTE while holding classifier hyperparameters fixed produced held-out ROC-AUC changes of only −0.0031 to +0.0029, with all ten paired confidence intervals again including zero. The reinterpreted SHAP pattern also held under both checks: full-column Spearman agreement declined (Phenotype A $\Delta = -0.036$ [−0.053, −0.013]; without SMOTE −0.069 [−0.088, −0.059]), whereas agreement over the shared 19 baseline columns did not decline (Phenotype A +0.029 [+0.008, +0.053]; without SMOTE +0.008 [−0.017, +0.017]; Appendix F).

Supplementary analyses were broadly consistent with the primary result. Brier scores and calibration were similar across arms, and 19 of 20 exploratory decision-curve net-benefit comparisons had intervals including zero; the exception, Bagged Decision Trees at a threshold of 0.5, favored the lifestyle arm (Appendix G). In feature-subset ablations, age alone reached held-out ROC-AUC 0.775–0.780 and the full baseline 0.804–0.810, whereas lifestyle alone reached 0.629–0.648 (Appendix J). Across ten training seeds on the original split, none of 50 paired lifestyle-minus-baseline intervals excluded zero, although Bagged Decision Trees showed a small positive difference in all ten seeds (+0.0001 to +0.0042); exploratory repeated random splits gave similar results (Appendix K).

In an exploratory analysis restricted to participants definitely not taking antihypertensive medication (n = 5,697; test set 566), with measured-BP Phenotype A as the outcome, all five models showed positive lifestyle-minus-baseline ROC-AUC differences (+0.0003 to +0.018). Unadjusted intervals excluded zero for Bagged Decision Trees (+0.018 [+0.002, +0.035]) and, narrowly, XGBoost (+0.012 [+0.00001, +0.024]; Appendix I). This analysis changes both the population and the outcome definition, reuses hyperparameters tuned for the primary phenotype in the full cohort, has a small test set, and involves five comparisons not adjusted for multiplicity; it is a sensitivity analysis, not an estimate of a treatment effect.

4. Discussion and Conclusion

In the full NHANES analytic cohort, the evaluated lifestyle block of physical activity, alcohol, smoking, and sleep information showed little detectable incremental discrimination beyond demographic, socioeconomic, and anthropometric predictors, across five model families and under both robustness checks. Exploratory analysis among definitely untreated adults yielded consistently positive, generally small AUC differences, with two nominal unadjusted intervals excluding zero; these findings merit confirmation in independently validated, appropriately tuned analyses. Because that analysis changes both the population and the outcome definition, it cannot distinguish among differences in treatment status, outcome definition, or population characteristics. None of this is evidence that lifestyle behaviors are unassociated with hypertension or clinically unimportant; established risk-factor associations and incremental predictive discrimination in a specific analytic cohort are different questions, and these results speak only to the latter.

Descriptive diagnostics on the development set help explain the small increment (Appendix H). The lifestyle variables are not duplicates of the baseline: their largest pairwise Spearman correlation with a baseline predictor was moderate ($|\rho| = 0.30$, vigorous activity vs. age), and the full baseline predicted each lifestyle variable only weakly. However, many lifestyle–outcome associations weakened substantially after accounting for baseline-predicted risk (e.g., vigorous activity, $\rho = -0.16$ to -0.05; former smoking, 0.13 to 0.01), with trouble sleeping a partial exception (0.14 to 0.08). This pattern is consistent with overlapping predictive information; it does not establish conditional independence or a causal explanation.

The full-column decline in cross-model SHAP agreement should not be read as a general deterioration of explanation agreement. Over the 19 baseline columns shared by both arms, the estimated change was small, with a confidence interval spanning zero, and a single shuffled-lifestyle control showed a similar full-column decline. This supports, but does

Table 2: Mean pairwise cross-model SHAP agreement over encoded columns, primary phenotype. "All" compares each condition over its full column set; "shared 19" restricts every condition to the 19 baseline columns. The shuffled row independently permutes each lifestyle predictor within the development and test sets (a single realization) and refits all models. Raw top-10 Jaccard values are not comparable across list lengths (chance level 0.365 for 19 columns, 0.167 for 36). Δ rows give the paired bootstrap difference (real lifestyle minus baseline) over the 300 explained participants; these intervals do not include model-refit variability.

		Spearman ρ		**Top-10 Jaccard**	
Condition	**Columns**	**All**	**Shared 19**	**All**	**Shared 19**
Baseline	19	0.873	0.873	0.821	0.821
Lifestyle-augmented (real)	36	0.806	0.863	0.699	0.761
Lifestyle-augmented (shuffled)	36	0.776	0.863	0.745	0.788
Δ real − baseline		−0.067	−0.010	−0.122	−0.061
95% CI		[−0.081, −0.049]	[−0.021, 0.010]	[−0.189, −0.045]	[−0.104, 0.030]

not prove, the interpretation that the full-column comparison is sensitive to adding low-importance columns that models order inconsistently. Real lifestyle columns nevertheless received more attribution than shuffled ones and entered some models' top 10 but not others, so the remaining disagreement is concentrated among the small lifestyle features rather than the baseline. The broader methodological point is that comparisons of attribution agreement across feature sets of different sizes require a matched comparison space; agreement over the full encoded space conflates changes in how models rank shared predictors with the addition of new predictors to rank. Aggregating attributions to predictor or domain level is a useful complement, since all five models agreed on the leading domains (Appendix F), although such aggregation can mask opposing within-group contributions. Huang and Huang (2023) similarly found that demographic factors dominated SHAP-based importance over a nutritional covariate set in the same NHANES cycle; this finding is a related but methodologically distinct observation, since theirs concerns within-model attribution rather than incremental ablation.

Several limitations bound these conclusions. The analysis uses a single NHANES cycle without external validation; testing whether the pattern holds in other NHANES cycles or in external cohorts is the main direction for future work. Estimates describe predictive performance within this analytic cohort rather than a survey-weighted, nationally representative population. Lifestyle here refers specifically to the activity, alcohol, smoking, and sleep block evaluated; diet and laboratory biomarkers were excluded by design, and a broader block may behave differently. Variables were selected from the study aims and the cited literature without a formal clinical expert review. Performance was not evaluated within sex, age, or race/ethnicity subgroups. The held-out test set was examined once with default hyperparameters before tuning, and all supplementary analyses reuse it, so the final evaluation is not fully independent of earlier modeling decisions (Section 2). The SMOTE sensitivity result reflects hyperparameters selected under the SMOTE-inclusive primary pipeline rather than independently retuned alternatives. Future work could also examine blood-pressure control among the 2,313 participants currently taking antihypertensive medication, which is a distinct outcome and population, and other modifiable domains such as diet. For this application, the baseline predictor set captured nearly as much discriminative information as the lifestyle-augmented set.

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Appendix A. Cohort, Phenotype, and Variable Specification

Hypertension was defined using the 130/80 mmHg threshold established by Whelton et al. (2018). Three definitions were computed from the same cohort: (A) measured systolic ≥130 mmHg or diastolic ≥80 mmHg; (B, primary) A or currently taking antihypertensive medication; (C) B or a self-reported hypertension diagnosis. Table 3 reports all three. Within Phenotype B's 4,311 positives, 3,408 met the measured-BP criterion and 903 were medication-only positives (normal measured BP, on medication) – confirming Phenotype B's medication clause meaningfully changes case status. Measured BP was averaged across all available oscillometric readings (up to three) per participant.

Table 3: Prevalence under all three phenotype definitions, N=8,024.

Phenotype	Positive	Prevalence
A (measured only)	3,408	42.5%
B (primary)	4,311	53.7%
C (broadest)	4,555	56.8%

Table 4 summarizes the 45 raw NHANES variables retained for phenotype construction, baseline predictors, and lifestyle-feature derivation, grouped by role. Lifestyle variables were curated from 26 raw questionnaire items into 11 derived features to avoid redundant skip-pattern-laden columns; e.g., a Y/N activity question and its conditional days/minutes follow-ups were collapsed into one weekly-minutes total, with a structural zero (not a missing value) recorded when the parent Y/N response was negative. The same structural-zero logic was applied to alcohol and smoking status.

Model predictors. The baseline arm comprised age (`RIDAGEYR`), sex (`is_female`), race/ethnicity (`RIDRETH3`), education (`DMDEDUC2`), family income-to-poverty ratio (`INDFMPIR`), weight (`BMXWT`), height (`BMXHT`), BMI (`BMXBMI`), and waist circumference (`BMXWAIST`). The lifestyle-augmented arm additionally included weekly vigorous activity (`total_vigorous_min_wk`), weekly moderate activity (`total_moderate_min_wk`), daily sedentary time (`sedentary_min_day`), drinking status (`drinking_status`), average drinks/day (`alq130_avg_drinks_filled`), smoking status (`smoking_status`), recent tobacco use (`smoked_recently`), weekday sleep duration (`SLD012`), weekend sleep duration (`SLD013`), trouble sleeping (`trouble_sleeping`), and daytime sleepiness (`SLQ120`).

Survey design variables (sample weights, primary sampling units, strata) were retained in the processed dataset but were not used as model predictors; all reported estimates describe predictive performance within this analytic cohort and are not weighted to be nationally representative.

Variable selection. The lifestyle block was prespecified rather than selected from the data. The study asks whether a block of commonly recommended modifiable behaviors (physical activity, alcohol, smoking, and sleep) adds value beyond a routine baseline; these domains follow the guideline and meta-analytic evidence cited in Section 1 (Whelton et al., 2018; Barone Gibbs et al., 2021; Cecchini et al., 2024; Hosseini et al., 2024; Virdis et al., 2010). Selecting features using the outcome on the same data would tune the block to the data it is tested on and bias the incremental comparison. Every variable in the nine NHANES modules used has a recorded inclusion or exclusion decision with a stated reason; for example, snoring and breathing-pause items were excluded as sleep-apnea screening rather than lifestyle habits, age-restricted and interview-logistics items were excluded, and raw bed and wake times were replaced by derived sleep duration. Questionnaire skip patterns were respected, so that a "no" to a parent question became a structural zero rather than a missing value. Variables were chosen from the study aims

Table 4: NHANES variables by category, source modules, and role.

Category	N vars	Source modules
Phenotype	10	P_BPQ, P_BPXO
Demographic/SES	5	P_DEMO
Anthropometric	4	P_BMX
Lifestyle (raw)	26	P_PAQ, P_ALQ, P_SLQ, P_SMQ, P_SMQRTU

and the cited literature without a formal clinical expert review. The full variable-by-variable record, including candidates for future expansion such as waist-to-hip ratio and diabetes status, is in the code repository.

Cohort characteristics. Table 5 describes the 8,024 phenotype-eligible participants overall and by hypertension status under Phenotypes B and A. These summaries are unweighted and describe the analytic cohort rather than the U.S. population.

Appendix B. Training and Hyperparameter Details

Each pipeline applied one-hot encoding to categorical features, median imputation, standardization (Logistic Regression only), and SMOTE (Chawla et al., 2002), implemented via an `imbalanced-learn` Pipeline (Lemaitre et al., 2016) so that all fitted steps were refit within each training fold, never on validation or test data. Models were implemented in scikit-learn (Pedregosa et al., 2012): Logistic Regression, Random Forest (Breiman, 2001), Gradient Boosting (Friedman, 2001), and Bagged Decision Trees (Breiman, 1996); XGBoost (Chen and Guestrin, 2016) used its native implementation. Hyperparameters were selected via 5-fold stratified cross-validation (ROC-AUC) on the fixed 90% development set (Kohavi, 1995): an exhaustive grid for Logistic Regression, and 25-candidate randomized search (Bergstra and Bengio, 2012) for the remaining four models, run independently for each of the two feature arms (Table 6). Selected **classifier hyperparameters** (Table 7) were frozen and reused, unmodified, for every downstream analysis (held-out evaluation, SHAP, both robustness checks) – no hyperparameter was re-selected using held-out test performance or any sensitivity-analysis result. Two robustness checks deliberately varied other pipeline elements while holding classifier hyperparameters fixed: the SMOTE sensitivity analysis (Appendix E) toggled SMOTE's presence, and the Phenotype A analysis changed the prediction target. Model-family comparisons are also sensitive to tuning: XGBoost required substantial hyperparameter search to reach parity with the other four models on this dataset size, so a comparison using library defaults alone would have misleadingly suggested a real model-family difference that tuning mostly closed.

Search budget and sensitivity. The 25-candidate budget for the tree models was fixed before the search was run, applied uniformly across the four tree-based model families so that tuning effort would not differ between them, and not changed in response to results; random search typically finds near-optimal configurations with modest budgets when only a few hyperparameters matter (Bergstra and Bengio, 2012). Table 8 summarizes the full search log. In every arm and model, several sampled configurations reached cross-validated ROC-AUC within 0.002 of the best, and the margin between the best and second-best configurations ($\leq$0.0005) was far smaller than the cross-validation standard deviation of the selected configuration (about 0.012). This indicates that the sampled candidates performed similarly; it does not establish that the search converged, and modest gains from a much larger search cannot be ruled out. As context only, the tuned tree models reached the same cross-validated ROC-AUC as the exhaustively tuned Logistic Regression (0.813 baseline, 0.815 lifestyle-augmented), and tuning substantially improved weak library defaults (e.g., XGBoost 0.782 to 0.813 and Bagged Decision Trees 0.791 to 0.813 in the baseline arm).

Appendix C. Complete Predictive Results

Table 9 reports 5-fold development CV performance (mean $\pm$ SD) for the frozen configurations above. Ta-

Table 5: Characteristics of the analytic cohort, overall and by hypertension status under the primary Phenotype B (measured ≥130/80 mmHg or antihypertensive medication) and Phenotype A (measured BP only). Summaries are unweighted and describe the analytic cohort, not the U.S. population. Percentages use non-missing values; "Missing" counts missing values in the overall cohort. Age is top-coded at 80 and the income-to-poverty ratio at 5 in the public NHANES release. Daytime sleepiness is the share reporting feeling overly sleepy often or almost always.

Characteristic	**Overall** $n = 8{,}024$	**B: yes** $n = 4{,}311$	**B: no** $n = 3{,}713$	**A: yes** $n = 3{,}408$	**A: no** $n = 4{,}616$	**Missing**
Demographic/SES						
Age, years, mean (SD)	49.4 (18.3)	57.7 (15.4)	39.8 (16.6)	56.2 (16.0)	44.3 (18.2)	0
Female, n (%)	4,049 (50.5)	2,042 (47.4)	2,007 (54.1)	1,594 (46.8)	2,455 (53.2)	0
Race/ethnicity, n (%)						0
Mexican American	899 (11.2)	375 (8.7)	524 (14.1)	309 (9.1)	590 (12.8)	
Other Hispanic	810 (10.1)	394 (9.1)	416 (11.2)	311 (9.1)	499 (10.8)	
Non-Hispanic White	2,842 (35.4)	1,540 (35.7)	1,302 (35.1)	1,166 (34.2)	1,676 (36.3)	
Non-Hispanic Black	2,147 (26.8)	1,362 (31.6)	785 (21.1)	1,103 (32.4)	1,044 (22.6)	
Non-Hispanic Asian	928 (11.6)	436 (10.1)	492 (13.3)	353 (10.4)	575 (12.5)	
Other/multiracial	398 (5.0)	204 (4.7)	194 (5.2)	166 (4.9)	232 (5.0)	
Education, n (%)						381
Less than 9th grade	518 (6.8)	324 (7.6)	194 (5.8)	263 (7.8)	255 (6.0)	
9th–11th grade	833 (10.9)	505 (11.8)	328 (9.8)	411 (12.2)	422 (9.9)	
High school/GED	1,858 (24.3)	1,103 (25.8)	755 (22.5)	866 (25.6)	992 (23.3)	
Some college/AA	2,533 (33.1)	1,403 (32.8)	1,130 (33.6)	1,107 (32.8)	1,426 (33.5)	
College graduate+	1,901 (24.9)	946 (22.1)	955 (28.4)	733 (21.7)	1,168 (27.4)	
Income-to-poverty ratio, med. [IQR]	2.19 [1.17, 4.17]	2.24 [1.21, 4.15]	2.15 [1.11, 4.22]	2.16 [1.18, 4.02]	2.22 [1.16, 4.35]	1,041
Anthropometric						
Weight, kg, mean (SD)	83.7 (23.0)	87.6 (23.6)	79.1 (21.4)	86.8 (23.3)	81.4 (22.5)	63
Height, cm, mean (SD)	167.0 (10.0)	167.1 (10.2)	167.0 (9.8)	167.2 (10.3)	166.9 (9.9)	71
BMI, kg/m^2, mean (SD)	29.9 (7.5)	31.2 (7.5)	28.3 (7.1)	30.9 (7.3)	29.1 (7.5)	79
Waist, cm, mean (SD)	100.5 (17.4)	105.3 (16.6)	95.1 (16.6)	104.4 (16.4)	97.8 (17.5)	244
Lifestyle						
Vigorous activity, min/wk, med. [IQR]	0 [0, 240]	0 [0, 180]	40 [0, 360]	0 [0, 180]	0 [0, 315]	2
Moderate activity, min/wk, med. [IQR]	180 [0, 690]	150 [0, 600]	225 [10, 840]	150 [0, 600]	180 [0, 720]	1
Sedentary, min/day, med. [IQR]	300 [180, 480]	300 [180, 480]	300 [180, 480]	300 [180, 480]	300 [180, 480]	57
Drinking status, n (%)						297
Current	5,469 (70.8)	2,758 (66.6)	2,711 (75.6)	2,210 (67.6)	3,259 (73.1)	
Former	1,497 (19.4)	1,006 (24.3)	491 (13.7)	740 (22.6)	757 (17.0)	
Never	761 (9.8)	375 (9.1)	386 (10.8)	321 (9.8)	440 (9.9)	
Drinks/day, med. [IQR]	1.0 [0.0, 2.0]	1.0 [0.0, 2.0]	1.0 [1.0, 3.0]	1.0 [0.0, 2.0]	1.0 [0.0, 2.0]	307
Smoking status, n (%)						4
Current	1,432 (17.9)	795 (18.5)	637 (17.2)	651 (19.1)	781 (16.9)	
Former	1,818 (22.7)	1,195 (27.7)	623 (16.8)	903 (26.5)	915 (19.8)	
Never	4,770 (59.5)	2,318 (53.8)	2,452 (66.1)	1,851 (54.4)	2,919 (63.3)	
Recent tobacco use, n (%)	1,844 (23.9)	975 (23.6)	869 (24.2)	796 (24.3)	1,048 (23.5)	295
Weekday sleep, h, mean (SD)	7.6 (1.7)	7.6 (1.7)	7.6 (1.6)	7.5 (1.7)	7.6 (1.7)	65
Weekend sleep, h, mean (SD)	8.3 (1.8)	8.1 (1.8)	8.4 (1.7)	8.1 (1.8)	8.4 (1.8)	66
Trouble sleeping, n (%)	2,290 (28.6)	1,471 (34.1)	819 (22.1)	1,062 (31.2)	1,228 (26.6)	5
Daytime sleepiness, n (%)	2,089 (26.1)	1,131 (26.3)	958 (25.8)	854 (25.1)	1,235 (26.8)	13

Table 6: Hyperparameter search spaces. LR used an exhaustive grid; all others used 25-candidate randomized search.

Model	Search space
Logistic Regression	$C \in \{0.01, 0.1, 1, 10, 100\}$
Random Forest	`n_estimators` $\in \{200, 300, 500\}$ `max_depth` $\in \{\text{None}, 5, 10, 20\}$ `min_samples_leaf` $\in \{1, 2, 5, 10\}$ `max_features` $\in \{\text{sqrt}, \text{log2}, 0.5\}$
Gradient Boosting	`n_estimators` $\in \{100, 200, 300\}$ `learning_rate` $\in \{.01, .05, .1, .2\}$ `max_depth` $\in \{2, 3, 4, 5\}$ `min_samples_leaf` $\in \{1, 5, 10, 20\}$
Bagged Decision Tree	`n_estimators` $\in \{200, 300, 500\}$ base `max_depth` $\in \{\text{None}, 5, 10, 20\}$ base `min_samples_leaf` $\in \{1, 2, 5, 10\}$ `max_samples` $\in \{.5, .7, 1.0\}$
XGBoost	`n_estimators` $\in \{100, 200, 300, 500\}$ `learning_rate` $\in \{.01, .05, .1, .2\}$ `max_depth` $\in \{3, 4, 5, 6, 8\}$ `subsample` $\in \{.6, .8, 1.0\}$ `colsample_bytree` $\in \{.6, .8, 1.0\}$ `min_child_weight` $\in \{1, 3, 5\}$

ble 10 reports held-out test performance with bootstrap 95% confidence intervals (Efron and Tibshirani, 1993). Table 11 reports the paired lifestyle-minus-baseline delta per model (main-text Figure 1 visualizes the ROC-AUC column of this table).

Appendix D. Interpretability Details

SHAP values (Lundberg and Lee, 2017) were computed with `TreeExplainer` (Lundberg et al., 2020) for Random Forest, Gradient Boosting, and XGBoost. `TreeExplainer` does not support `BaggingClassifier`; a model-agnostic permutation explainer was substituted for that model only. Cross-model agreement (Table 2) was assessed over the post-encoding feature space (19 baseline, 36 lifestyle-augmented columns) and, for the matched comparison, over the 19 baseline columns shared by both arms (Appendix F). The illustrative top-feature visualization (Figure 2) instead aggregates one-hot-encoded categorical predictors (e.g., race/ethnicity) back to their source variable, for interpretability: SHAP values for a predictor's encoded columns were summed within each sample before absolute values were averaged across samples. These are two distinct units of analysis and are not directly comparable. Cross-model agreement was quantified two ways, following Krishna et al. (2022)'s characterization of explanation disagreement across models: top-10 feature-membership Jaccard overlap, and Spearman rank correlation over the complete importance ordering (Table 2). ρ magnitudes are reported descriptively; p-values are not treated as a formal hypothesis test given the small, non-independent feature set.

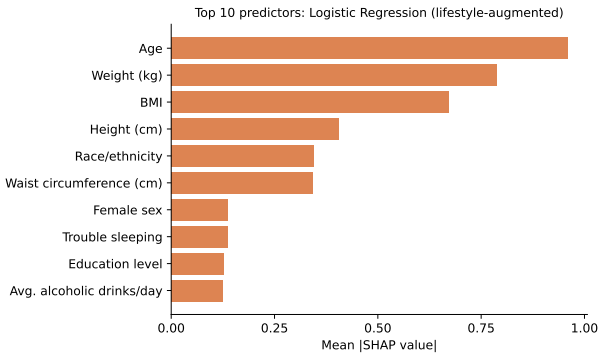


Figure 2: Top-10 predictor-level SHAP importance for Logistic Regression, which achieved the highest development-set CV ROC-AUC in the lifestyle-augmented arm.

Appendix E. Robustness and Sensitivity Analyses

Phenotype A. All ten frozen hyperparameter configurations were refit under Phenotype A (Table 12), using the identical 90/10 participant split as the primary analysis (stratified on Phenotype B) so that only the label, not cohort composition, varied. The split is not meaningfully imbalanced for Phenotype A: its prevalence is 42.6% in the development set (3,074/7,221) and 41.6% in the test set (334/803), against 42.5% overall.

SMOTE sensitivity. Using the same frozen hyperparameters, each model was refit with SMOTE included and excluded (all other pipeline steps identical). Table 13 reports the 10 paired SMOTE-minus-no-SMOTE comparisons across five models and two feature arms. All 10 paired-bootstrap 95% AUROC confidence intervals included zero.

Table 7: Selected hyperparameters, frozen for all downstream analyses.

Arm	Model	Selected hyperparameters
Baseline	Logistic Regression	C=10
Baseline	Random Forest	n_estimators=200, min_samples_leaf=10, max_features=0.5, max_depth=5
Baseline	Gradient Boosting	n_estimators=100, min_samples_leaf=20, max_depth=3, learning_rate=0.05
Baseline	Bagged Decision Tree	n_estimators=500, max_samples=0.5, base min_samples_leaf=10, base max_depth=5
Baseline	XGBoost	subsample=1.0, n_estimators=300, min_child_weight=5, max_depth=4, learning_rate=0.01, colsample_bytree=0.8
Lifestyle-augmented	Logistic Regression	C=100
Lifestyle-augmented	Random Forest	n_estimators=500, min_samples_leaf=2, max_features=0.5, max_depth=5
Lifestyle-augmented	Gradient Boosting	n_estimators=100, min_samples_leaf=10, max_depth=4, learning_rate=0.05
Lifestyle-augmented	Bagged Decision Tree	n_estimators=300, max_samples=0.5, base min_samples_leaf=10, base max_depth=10
Lifestyle-augmented	XGBoost	subsample=1.0, n_estimators=300, min_child_weight=5, max_depth=4, learning_rate=0.01, colsample_bytree=0.8

Table 8: Hyperparameter search summary, 5-fold development-set CV ROC-AUC. "Default" is library-default CV ROC-AUC before tuning; "Best" is the selected configuration; "Gap to 2nd" is its margin over the next-best candidate; "Within 0.002" counts candidates within 0.002 of the best; "SD of best" is the CV fold standard deviation of the selected configuration. Logistic Regression used an exhaustive five-value grid; tree models used 25 randomized candidates.

Arm	Model	Cand.	Default	Best	Gap to 2nd	Within 0.002	SD of best
Baseline	Logistic Regression	5	0.813	0.813	0.0000	5/5	0.013
	Random Forest	25	0.796	0.812	0.0001	10/25	0.012
	Gradient Boosting	25	0.810	0.812	0.0001	4/25	0.012
	Bagged Decision Tree	25	0.791	0.813	0.0001	8/25	0.012
	XGBoost	25	0.782	0.813	0.0003	5/25	0.012
Lifestyle-augmented	Logistic Regression	5	0.815	0.815	0.0000	5/5	0.010
	Random Forest	25	0.804	0.812	0.0000	11/25	0.013
	Gradient Boosting	25	0.814	0.815	0.0005	5/25	0.013
	Bagged Decision Tree	25	0.800	0.813	0.0004	8/25	0.012
	XGBoost	25	0.785	0.815	0.0002	4/25	0.013

Table 9: Development-set 5-fold CV performance, frozen hyperparameters.

Arm	Model	ROC-AUC	SD	PR-AUC	SD
Baseline	Logistic Regression	0.8128	0.0125	0.8102	0.0171
Baseline	Random Forest	0.8122	0.0123	0.8113	0.0126
Baseline	Gradient Boosting	0.8118	0.0120	0.8097	0.0127
Baseline	Bagged Decision Tree	0.8125	0.0122	0.8114	0.0122
Baseline	XGBoost	0.8130	0.0124	0.8125	0.0131
Lifestyle-augmented	Logistic Regression	0.8154	0.0100	0.8135	0.0093
Lifestyle-augmented	Random Forest	0.8117	0.0128	0.8102	0.0134
Lifestyle-augmented	Gradient Boosting	0.8153	0.0130	0.8114	0.0145
Lifestyle-augmented	Bagged Decision Tree	0.8131	0.0122	0.8078	0.0130
Lifestyle-augmented	XGBoost	0.8150	0.0128	0.8116	0.0140

Table 10: Held-out test performance with bootstrap 95% confidence intervals.

Arm	Model	ROC-AUC [95% CI]	PR-AUC [95% CI]
Baseline	Logistic Regression	0.809 [0.778, 0.838]	0.798 [0.755, 0.841]
Baseline	Random Forest	0.804 [0.773, 0.834]	0.792 [0.747, 0.837]
Baseline	Gradient Boosting	0.809 [0.779, 0.839]	0.802 [0.760, 0.844]
Baseline	Bagged Decision Tree	0.807 [0.776, 0.837]	0.796 [0.753, 0.838]
Baseline	XGBoost	0.810 [0.779, 0.840]	0.796 [0.754, 0.840]
Lifestyle-augmented	Logistic Regression	0.807 [0.777, 0.837]	0.791 [0.748, 0.834]
Lifestyle-augmented	Random Forest	0.804 [0.772, 0.832]	0.791 [0.746, 0.835]
Lifestyle-augmented	Gradient Boosting	0.808 [0.779, 0.838]	0.798 [0.756, 0.840]
Lifestyle-augmented	Bagged Decision Tree	0.807 [0.778, 0.838]	0.794 [0.751, 0.838]
Lifestyle-augmented	XGBoost	0.807 [0.777, 0.837]	0.795 [0.754, 0.838]

Table 11: Paired bootstrap deltas (lifestyle-augmented minus baseline), held-out test.

Model	Δ ROC-AUC [95% CI]	Δ PR-AUC [95% CI]
Logistic Regression	−0.0018 [−0.0083, 0.0052]	−0.0072 [−0.0187, 0.0060]
Random Forest	−0.0009 [−0.0044, 0.0028]	−0.0010 [−0.0070, 0.0052]
Gradient Boosting	−0.0018 [−0.0085, 0.0054]	−0.0044 [−0.0159, 0.0077]
Bagged Decision Tree	0.0002 [−0.0058, 0.0061]	−0.0019 [−0.0117, 0.0081]
XGBoost	−0.0028 [−0.0075, 0.0026]	−0.0007 [−0.0080, 0.0077]

Table 12: Phenotype A paired deltas (lifestyle-augmented minus baseline), held-out test.

Model	Δ ROC-AUC [95% CI]
Logistic Regression	0.0009 [−0.0090, 0.0108]
Random Forest	0.0021 [−0.0052, 0.0088]
Gradient Boosting	−0.0006 [−0.0115, 0.0093]
Bagged Decision Tree	0.0086 [−0.0020, 0.0199]
XGBoost	0.0038 [−0.0047, 0.0120]

Table 13: SMOTE-minus-no-SMOTE paired deltas with 95% CIs, held-out test, frozen hyperparameters. "Lifestyle" is the lifestyle-augmented arm.

Arm	Model	Δ ROC-AUC [95% CI]
Baseline	Logistic Regression	0.0002 [−0.0010, 0.0013]
	Random Forest	0.0029 [−0.0005, 0.0063]
	Gradient Boosting	0.0006 [−0.0024, 0.0036]
	Bagged Decision Tree	0.0012 [−0.0016, 0.0041]
	XGBoost	0.0005 [−0.0024, 0.0032]
Lifestyle	Logistic Regression	−0.0006 [−0.0018, 0.0008]
	Random Forest	0.0021 [−0.0012, 0.0058]
	Gradient Boosting	−0.0023 [−0.0069, 0.0020]
	Bagged Decision Tree	−0.0006 [−0.0048, 0.0035]
	XGBoost	−0.0031 [−0.0064, 0.0001]

Appendix F. SHAP Agreement Controls

These controls test whether the lower full-column SHAP agreement in the lifestyle-augmented arm (Table 2) reflects the lifestyle features themselves or the larger number of columns being ranked (36 vs. 19). The matched-space, predictor-level, and domain-level views reuse the SHAP values already computed for the same 300 test participants, so no model is refit; agreement is the mean over the ten model pairs, and paired bootstrap intervals resample the explained participants and therefore exclude model-refit variability.

Matched comparison space. The lifestyle arm's agreement is recomputed over only the 19 baseline columns that the baseline arm ranks. Table 14 reports this comparison for the primary phenotype, Phenotype A, and the no-SMOTE refit. In all three settings, full-column Spearman agreement declined, while agreement over the shared 19 columns did not decline. Under Phenotype A the shared-column interval lay above zero (+0.029 [+0.008, +0.053]).

Shuffled-lifestyle control. Each lifestyle predictor was permuted independently within the development and test sets, preserving the 36-column feature space and each predictor's marginal distribution while disrupting its observed associations with other predictors and with the outcome. All five models were refit with the frozen hyperparameters and SHAP was recomputed. This is a single permutation realization, and independent permutation can create unrealistic combinations of predictor values, so it supports but does not prove the interpretation in Section 3. Shuffled columns received 5.3% of total mean |SHAP|, compared with 11.3% for the real lifestyle columns.

Coarser units. When one-hot columns are summed back to source predictors, mean pairwise Spearman agreement is 0.884 in the lifestyle arm (20 predictors) vs. 0.770 in the baseline arm (9 predictors); the lists differ in length, so this comparison is descriptive only. At the domain level (Table 15), all five models rank demographic/SES first and anthropometric second, and the four lifestyle domains together account for about 13% of net grouped SHAP magnitude. Disagreement is concentrated among the small lifestyle domains, whose order varies across models. Under Phenotype A, domain-level agreement was 0.84, with the same top two domains.

Table 14: Cross-model SHAP agreement by setting (mean pairwise over five models; Base. = baseline arm, Life. = lifestyle-augmented arm). "All" uses each arm's full encoded column set (19 vs. 36); "shared" restricts the lifestyle arm to the 19 baseline columns, so the baseline value is unchanged. Jaccard is top-10 overlap. Δ is lifestyle minus baseline; 95% CIs are paired bootstraps over the 300 explained participants and do not include model-refit variability. Chance-level top-10 Jaccard is 0.365 for 19 columns and 0.167 for 36.

Setting	Metric	Base.	Life., all	Δ all [95% CI]	Life., shared	Δ shared [95% CI]
Primary	Spearman ρ	0.873	0.806	−0.067 [−0.081, −0.049]	0.863	−0.010 [−0.021, +0.010]
	Jaccard	0.821	0.699	−0.122 [−0.189, −0.045]	0.761	−0.061 [−0.104, +0.030]
Phenotype A	Spearman ρ	0.830	0.794	−0.036 [−0.053, −0.013]	0.858	+0.029 [+0.008, +0.053]
	Jaccard	0.686	0.668	−0.018 [−0.074, −0.007]	0.742	+0.056 [0.000, +0.126]
No SMOTE	Spearman ρ	0.864	0.795	−0.069 [−0.088, −0.059]	0.873	+0.008 [−0.017, +0.017]
	Jaccard	0.773	0.702	−0.070 [−0.099, −0.003]	0.758	−0.015 [−0.018, +0.030]

Appendix G. Calibration and Decision-Curve Analysis

Table 15: Domain-level SHAP importance, lifestyle-augmented arm, primary phenotype. Per-sample signed SHAP values are summed within each domain before absolute values are averaged over samples (the net grouped contribution, as in Figure 2); opposing attributions within a domain can cancel. "Mean share" averages each domain's share of net grouped SHAP magnitude across the five models; remaining columns give each model's domain rank. Mean pairwise Spearman agreement over the six domains is 0.903 [+0.874, +0.931].

Domain	Share	LR	RF	GB	BDT	XGB
Demographic/SES	63.1%	1	1	1	1	1
Anthropometric	24.3%	2	2	2	2	2
Sleep	4.9%	3	4	3	3	3
Smoking	3.0%	5	3	4	4	4
Alcohol	2.4%	4	6	6	6	6
Activity	2.3%	6	5	5	5	5

The frozen models were evaluated on the same held-out test set as the main analysis (Table 16; Figure 3). Brier-score differences were negligible, and all five paired intervals included zero. Both arms slightly underestimate risk on average: mean predicted risk was 0.50–0.52 against an observed rate of 0.537, consistent with the positive calibration-in-the-large values. Separately, calibration slopes at or somewhat above 1 indicate that predictions are, if anything, slightly insufficiently spread out rather than overconfident. Under the primary phenotype the negatives are the minority class, so SMOTE oversamples them during training; this may contribute to the downward shift seen in both arms.

An exploratory decision-curve analysis (Vickers and Elkin, 2006) compared net benefit across risk thresholds from 0.05 to 0.80 (Figure 4). The two arms' curves lie close together. Of the 20 paired comparisons at four thresholds (Table 17), 19 intervals included zero; the exception, Bagged Decision Trees at 0.5 (+0.020 [+0.005, +0.036]), is isolated among correlated, unadjusted comparisons, and the other four models' differences at that threshold differ in sign. These thresholds are not clinically validated decision points for this cross-sectional classification task, so the curves are not interpreted as evidence of deployable clinical utility for either arm.

Table 16: Calibration on the held-out test set, frozen models (baseline / lifestyle-augmented). Calibration-in-the-large is the intercept with the slope fixed at 1; calibration slope is estimated separately with a free intercept. Both are point estimates. Observed prevalence is 0.537. Δ Brier is lifestyle minus baseline with a paired bootstrap 95% CI (lower Brier is better).

Model	**Brier**	**Cal. slope**	**CITL**	**Mean pred.**	**Δ Brier [95% CI]**
Logistic Regression	0.178 / 0.178	1.01 / 0.99	+0.19 / +0.18	0.503 / 0.504	0.0000 [−0.0031, +0.0030]
Random Forest	0.179 / 0.179	1.13 / 1.10	+0.17 / +0.16	0.505 / 0.506	−0.0001 [−0.0016, +0.0013]
Gradient Boosting	0.176 / 0.176	1.11 / 1.05	+0.13 / +0.09	0.513 / 0.520	−0.0003 [−0.0034, +0.0028]
Bagged Decision Tree	0.178 / 0.177	1.05 / 1.00	+0.17 / +0.15	0.505 / 0.510	−0.0009 [−0.0036, +0.0018]
XGBoost	0.177 / 0.177	1.21 / 1.19	+0.14 / +0.13	0.511 / 0.513	+0.0005 [−0.0017, +0.0024]

Table 17: Exploratory decision-curve analysis: paired lifestyle-minus-baseline differences in net benefit at four risk thresholds p_t, held-out test set, with paired bootstrap 95% CIs. These 20 comparisons are correlated and unadjusted; the single interval excluding zero is in bold. Thresholds are illustrative, not clinically validated decision points.

Model	$p_t = 0.2$	$p_t = 0.3$	$p_t = 0.4$	$p_t = 0.5$
Logistic Regression	+0.002 [−0.005, +0.009]	−0.003 [−0.014, +0.009]	−0.002 [−0.014, +0.011]	+0.006 [−0.006, +0.020]
Random Forest	+0.001 [−0.006, +0.007]	+0.002 [−0.003, +0.007]	+0.006 [−0.0004, +0.013]	+0.006 [−0.004, +0.017]
Gradient Boosting	−0.002 [−0.007, +0.003]	−0.009 [−0.019, +0.001]	+0.001 [−0.012, +0.012]	−0.002 [−0.016, +0.011]
Bagged Decision Tree	−0.004 [−0.010, +0.002]	+0.002 [−0.008, +0.012]	+0.003 [−0.008, +0.015]	**+0.020 [+0.005, +0.036]**
XGBoost	+0.001 [−0.003, +0.005]	−0.006 [−0.015, +0.002]	+0.003 [−0.007, +0.015]	−0.004 [−0.019, +0.011]

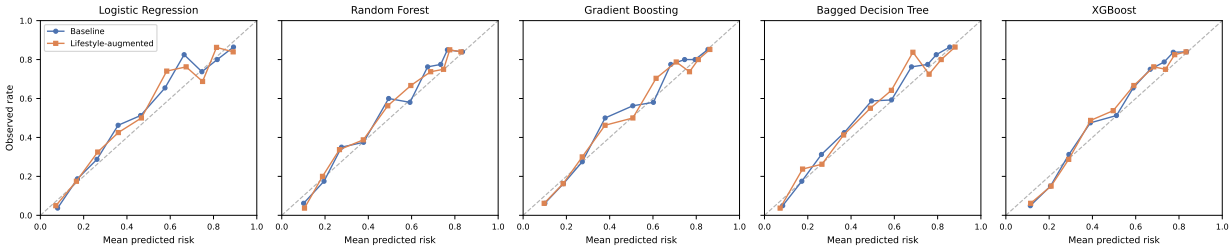


Figure 3: Reliability diagrams on the held-out test set, baseline vs. lifestyle-augmented arm, frozen models. The dashed line is perfect calibration.

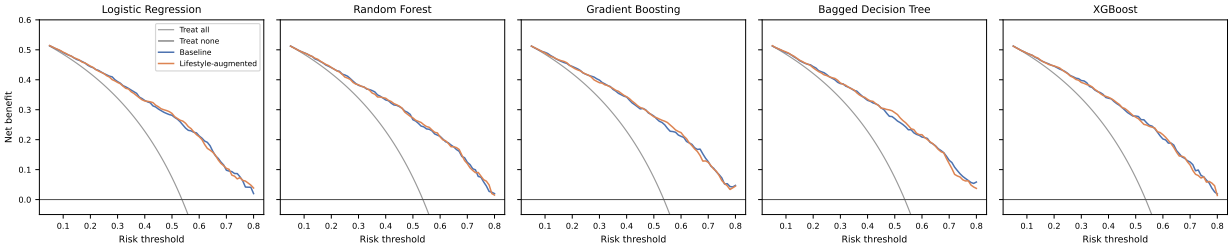


Figure 4: Exploratory decision curves on the held-out test set, baseline vs. lifestyle-augmented arm, with treat-all and treat-none references. Thresholds are illustrative, not validated decision points.

Appendix H. Lifestyle–Baseline Redundancy Diagnostics

These diagnostics use the development set only; the test set was not used. Each statistic uses participants with non-missing values for the variables involved, so sample sizes differ slightly across variables. Pairwise Spearman correlations between lifestyle variables and baseline predictors were at most moderate (Figure 5): the largest involved age (vigorous activity −0.30; drinks/day −0.28; former smoking and former drinking +0.26 each), and against BMI or waist circumference every $|\rho| \leq 0.17$. When all baseline predictors were used jointly to predict each lifestyle variable, cross-validated R^2 was at most 0.11 for continuous measures and ROC-AUC was 0.63–0.76 for binary indicators (Table 18).

To describe overlap with the outcome, each lifestyle variable was also correlated with the outcome residual, defined as Phenotype B minus the out-of-fold probability from a baseline-only model. Several associations weakened substantially, for example vigorous activity (−0.16 to −0.05), former drinking (+0.13 to +0.02), former smoking (+0.13 to +0.01), and moderate activity (−0.10 to −0.01); trouble sleeping was a partial exception (+0.14 to +0.08). This is consistent with overlapping predictive information between the lifestyle block and the full baseline, but the analysis is descriptive and does not establish conditional independence or a causal explanation.

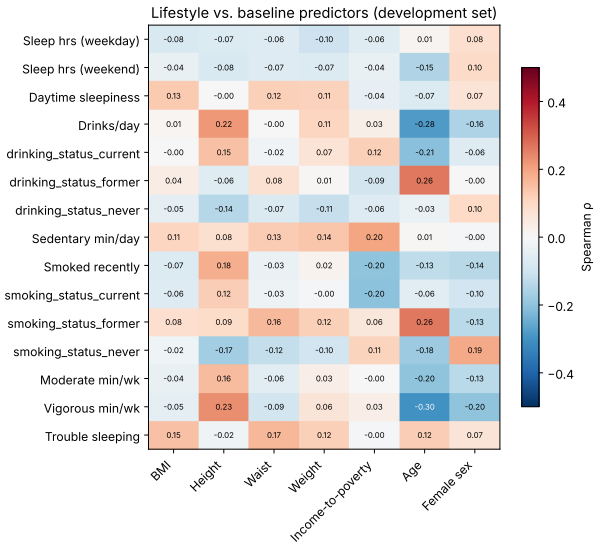


Figure 5: Pairwise Spearman correlations between lifestyle variables and numeric or binary baseline predictors, development set. Race/ethnicity and education indicators are omitted from the figure but included in Table 18's joint prediction.

Appendix I. Untreated-Adult Subgroup

This exploratory analysis was restricted to participants definitely not taking antihypertensive medi-

Table 18: Lifestyle–baseline redundancy diagnostics, development set only. "ρ with outcome" is the Spearman correlation with Phenotype B; "ρ with residual" is the correlation with the outcome minus the out-of-fold probability from a baseline-only model. "Predicted from baseline" is the 5-fold cross-validated performance (mean ± SD) of predicting the lifestyle variable from all baseline predictors (R^2 for continuous, ROC-AUC for binary indicators). Each statistic uses participants with non-missing values. These are descriptive and do not establish conditional independence or causation.

Lifestyle variable	**ρ with outcome**	**ρ with residual**	**Predicted from baseline**
Vigorous activity (min/wk)	−0.16	−0.05	R^2 0.08 ± 0.03
Trouble sleeping	+0.14	+0.08	AUC 0.63 ± 0.02
Drinking status: former	+0.13	+0.02	AUC 0.68 ± 0.02
Smoking status: never	−0.13	−0.04	AUC 0.73 ± 0.01
Smoking status: former	+0.13	+0.01	AUC 0.71 ± 0.01
Drinking status: current	−0.10	−0.01	AUC 0.69 ± 0.01
Moderate activity (min/wk)	−0.10	−0.01	R^2 0.04 ± 0.02
Average drinks/day	−0.10	+0.01	R^2 0.11 ± 0.02
Weekend sleep duration	−0.09	−0.02	R^2 −0.01 ± 0.01
Sedentary time (min/day)	+0.04	+0.01	R^2 0.07 ± 0.02
Drinking status: never	−0.03	−0.01	AUC 0.70 ± 0.02
Smoking status: current	+0.02	+0.04	AUC 0.76 ± 0.02
Weekday sleep duration	−0.02	0.00	R^2 −0.02 ± 0.02
Recent tobacco use	−0.01	+0.03	AUC 0.75 ± 0.01
Daytime sleepiness	0.00	+0.01	R^2 0.02 ± 0.02

cation according to the questionnaire skip pattern. The 14 participants with unknown treatment status were excluded rather than assumed untreated, leaving n = 5,697 (development 5,131; test 566). Because the medication clause of Phenotype B no longer applies, the outcome is measured-BP Phenotype A (prevalence about 34%). Hyperparameters are the frozen values tuned for the primary phenotype in the full cohort and were not retuned.

All five point estimates favored lifestyle augmentation (Table 19). Unadjusted intervals excluded zero for Bagged Decision Trees and, narrowly, XGBoost (lower bound +0.00001); the other three included zero. The subgroup test set is small, and the five comparisons are not adjusted for multiplicity. Because the analysis changes both the population and the outcome definition, the pattern may reflect differences in treatment status, in the outcome definition, or in the characteristics of the restricted population, and this analysis cannot distinguish among them. It is a sensitivity analysis, not an estimate of a treatment effect.

Appendix J. Feature-Subset Ablations

Table 20 reports held-out ROC-AUC on the original test set for nested and single-block predictor subsets. Each subset reuses the frozen hyperparameters of the corresponding arm and was not retuned, so the smaller subsets may be slightly disadvantaged. Age alone recovers most of the baseline's discrimination, and age plus BMI recovers nearly all of it. Adding the anthropometric block to demographic/SES predictors raised ROC-AUC by +0.009 to +0.014 across models; the paired interval excluded zero for Logistic Regression (+0.014 [+0.0001, +0.028]) and included zero for the other four. Lifestyle predictors alone discriminated above chance (0.629–0.648) but less well than age alone, which is not by itself evidence of clinical usefulness. Adding lifestyle to the full baseline showed no consistent detectable improvement in held-out ROC-AUC (−0.003 to 0.000).

Appendix K. Training-Seed and Repeated-Split Variability

The primary analysis uses a single split and seed, so two checks examine how much the lifestyle-

Table 19: Exploratory untreated-adult subgroup: participants definitely not taking antihypertensive medication, measured-BP Phenotype A outcome (development $n = 5{,}131$, prevalence 35.0%; test $n = 566$, prevalence 34.3%). Hyperparameters are the frozen primary-analysis values, not retuned. Δ is lifestyle minus baseline with a paired bootstrap 95% CI; intervals excluding zero are in bold. The five comparisons are not adjusted for multiplicity.

Model	**Baseline AUC [95% CI]**	**Lifestyle AUC [95% CI]**	**Δ [95% CI]**
Logistic Regression	0.717 [0.675, 0.756]	0.718 [0.673, 0.758]	+0.0003 [−0.0142, +0.0125]
Random Forest	0.702 [0.657, 0.748]	0.710 [0.665, 0.753]	+0.0078 [−0.0068, +0.0216]
Gradient Boosting	0.714 [0.669, 0.756]	0.720 [0.677, 0.763]	+0.0060 [−0.0091, +0.0221]
Bagged Decision Tree	0.700 [0.655, 0.746]	0.719 [0.675, 0.760]	**+0.0181 [+0.0020, +0.0348]**
XGBoost	0.706 [0.660, 0.750]	0.718 [0.674, 0.760]	**+0.0119 [+0.00001, +0.0241]**

Table 20: Feature-subset ablations: held-out ROC-AUC on the original test set by predictor subset (k = number of source predictors) and model. Subsets reuse the frozen hyperparameters of the corresponding arm (baseline arm for baseline subsets, lifestyle-augmented arm for subsets including lifestyle) and were not retuned, so they may be slightly disadvantaged. The last column gives the range across the five models of the paired difference from the full baseline.

Predictors	k	**LR**	**RF**	**GB**	**BDT**	**XGB**	**Δ vs. full baseline**
Age only	1	0.780	0.775	0.776	0.775	0.775	−0.035 to −0.029
Age + BMI	2	0.796	0.795	0.793	0.797	0.792	−0.018 to −0.010
Demographic/SES only	5	0.795	0.793	0.800	0.798	0.801	−0.014 to −0.009
Anthropometric only	4	0.681	0.685	0.686	0.686	0.685	−0.127 to −0.119
Full baseline	9	0.809	0.804	0.809	0.807	0.810	—
Lifestyle only	11	0.629	0.642	0.640	0.638	0.648	−0.180 to −0.162
Baseline + lifestyle	20	0.807	0.804	0.808	0.807	0.807	−0.003 to 0.000

Table 21: Variability of the lifestyle-minus-baseline held-out ROC-AUC difference over ten seeds. Training seeds keep the original development/test split and vary only the model and SMOTE random seeds. Repeated splits redraw the stratified 90/10 partition with ten seeds (including the original); because hyperparameters were tuned on the original development set, part of each new test set contributed to tuning, so these results describe split sensitivity conditional on that tuning, not independent validation. Each comparison has its own paired bootstrap 95% CI; the last column counts intervals excluding zero.

Model	**Mean AUC (base / life)**	**Mean Δ**	**SD**	**Range**	**$\Delta > 0$**	**CI excl. 0**
Training seeds, original split						
Logistic Regression	0.809 / 0.807	−0.0012	0.0002	−0.0015 to −0.0009	0/10	0/10
Random Forest	0.803 / 0.804	+0.0004	0.0010	−0.0007 to +0.0018	5/10	0/10
Gradient Boosting	0.809 / 0.810	+0.0012	0.0016	−0.0008 to +0.0047	8/10	0/10
Bagged Decision Tree	0.806 / 0.808	+0.0026	0.0014	+0.0001 to +0.0042	10/10	0/10
XGBoost	0.809 / 0.809	0.0000	0.0017	−0.0036 to +0.0026	6/10	0/10
Repeated random 90/10 splits (exploratory)						
Logistic Regression	0.814 / 0.816	+0.0022	0.0023	−0.0018 to +0.0051	8/10	0/10
Random Forest	0.814 / 0.813	−0.0009	0.0017	−0.0028 to +0.0023	2/10	0/10
Gradient Boosting	0.816 / 0.817	+0.0011	0.0022	−0.0018 to +0.0043	6/10	0/10
Bagged Decision Tree	0.814 / 0.815	+0.0004	0.0017	−0.0024 to +0.0023	7/10	0/10
XGBoost	0.816 / 0.816	+0.0002	0.0021	−0.0028 to +0.0046	6/10	1/10

minus-baseline difference varies (Table 21). First, keeping the original development/test split fixed, both arms were refit with ten different model and SMOTE random seeds. None of the 50 seed-by-model paired intervals excluded zero, and mean differences ranged from −0.0012 (Logistic Regression) to +0.0026 (Bagged Decision Trees). Bagged Decision Trees showed a positive difference in all ten seeds (+0.0001 to +0.0042), consistent in sign but small in size.

Second, as an exploratory check, the stratified 90/10 partition was redrawn with ten seeds, including the original. Hyperparameters were tuned on the original development set, so for the new seeds part of each new test set contributed to tuning; this is split sensitivity conditional on the original tuning, not independent validation. Mean differences ranged from −0.0009 to +0.0022, and one of 50 paired intervals excluded zero (XGBoost, seed 7: +0.0046 [+0.0006, +0.0089]). Mean AUCs in both arms were somewhat higher on the new splits than on the original test split (about 0.814–0.816 vs. 0.804–0.810), which may reflect partition-to-partition variation, the tuning overlap, or both.

Appendix L. Reproducibility and Computational Details

All code, result tables, figures, and raw SHAP arrays are available at https://github.com/urielbenymon/nhanes-lifestyle-hypertension. The repository provides the full execution order and maps every reported table and figure to the script and output file that produce it. In brief, the main pipeline downloads the NHANES files from CDC, builds the cohort and phenotypes, tunes and freezes the models on the development set, and evaluates them on the held-out test set; the robustness checks and all supplementary analyses then reuse the frozen hyperparameters, and the appendix tables are formatted directly from the saved result tables. The library-default (untuned) results cited in Section 2 are saved separately in the repository. A single random seed (42) was used throughout for splitting, cross-validation folds, SMOTE, and all stochastic model components, except in the training-seed, repeated-split, and shuffled-lifestyle analyses, which vary seeds by design. Table 22 reports wall-clock hyperparameter-search time per model (the dominant cost); all other stages completed in seconds to low minutes. Analyses used Python 3.14 with pandas 3.0.3, numpy 2.4.6, scikit-learn 1.9.0, xgboost 3.4.1, imbalanced-learn 0.14.2, shap 0.52.0, scipy 1.18.1, and matplotlib 3.11.1. Experiments were run on a Windows 11 system with an AMD Ryzen Z1 Extreme processor (3.30 GHz) and 16 GB of installed RAM (12.7 GB usable). No dedicated GPU was used.

Table 22: Hyperparameter search wall-clock time per model (both arms combined).

Model	Total search time
Logistic Regression	3.7s
Random Forest	188.8s
Gradient Boosting	619.7s
Bagged Decision Tree	213.2s
XGBoost	116.5s

NHANES data were obtained from https://wwwn.cdc.gov/nchs/nhanes/continuousnhanes/default.aspx?Cycle=2017-2020.

Appendix M. Ethics, Scope, and Responsible-Use Notes

This is a secondary analysis of publicly available, de-identified NHANES data; IRB status is stated in the Institutional Review Board statement accompanying this submission. No clinical deployment is proposed or validated. Reported associations describe predictive performance within this analytic cohort; survey weights, primary sampling units, and strata were not used, so estimates should not be interpreted as population-representative or causal (Appendix A). Performance was not evaluated within sex, age, or race/ethnicity subgroups, and fairness properties were not assessed; this is noted as a limitation rather than an assumption of uniform performance across demographic groups. The only subgroup analysis, restricted to definitely untreated adults (Appendix I), is an exploratory sensitivity analysis that changes both the population and the outcome definition; it is not a demographic subgroup or fairness evaluation.